

WASP-12b

R-0344

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231207 · created 2026-07-18T08:02:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460285.8612 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.012
Transit depth (Rp/Rs)^2	1.25 +/- 0.28 [%]
Orbital Inclination (inc)	86.79 +/- 1.09
Airmass coefficient 1 (a1)	0.9812 +/- 0.0074
Airmass coefficient 2 (a2)	0.017 +/- 0.034
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#2 - [288, 255]
Optimal Aperture	4.71
Optimal Annulus	13.19
Transit Duration (day)	0.1285 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.012 vs 0.1178 (deviation 0.28σ)
Duration fit vs published	0.1285 d vs 0.125 d (deviation 1.08σ)
Mid-time O–C	1.9 min
Depth SNR	5.5
Residual scatter	0.76 %

Light curve

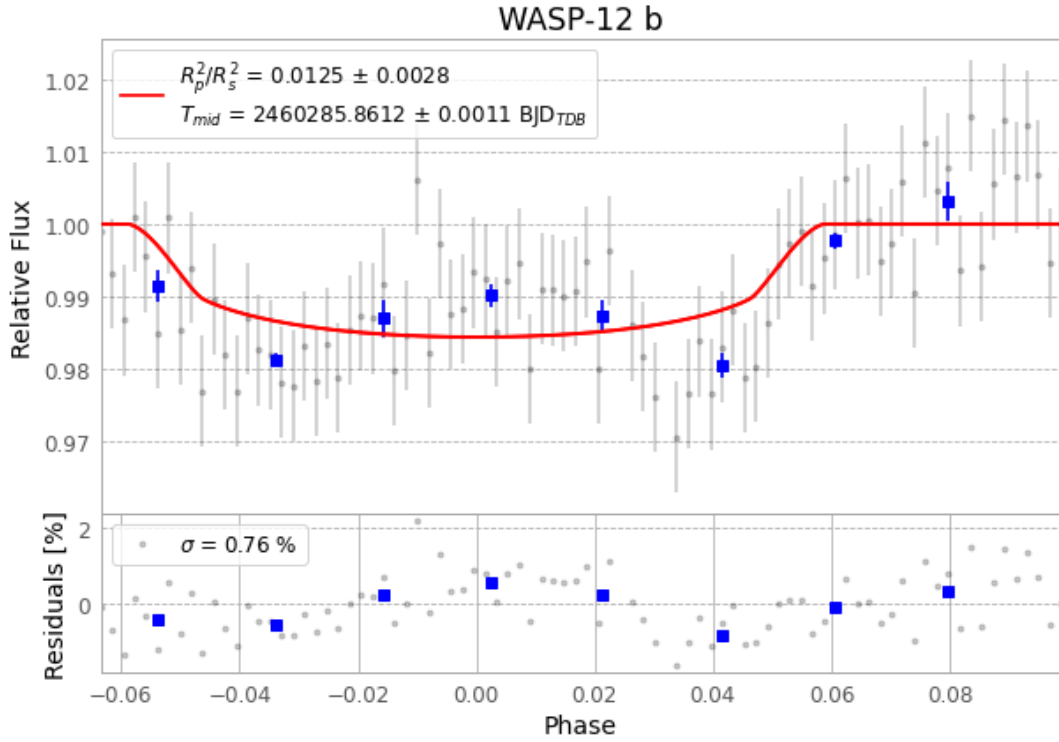


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-06.png (SHA-256 653e2ceed4dfb694...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 24e85c73d5e80b4c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

86 FITS frames (56 MB), WASP-12231207065115.FITS → WASP-12231207110615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231207.log (50d586806632...), FinalLightCurve_WASP-12 b_2023-12-06.png (653e2ceed4df...), FinalLightCurve_WASP-12 b_2023-12-06.csv (d8926f73b43a...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 08:02Z.